# Hypothesis tests for multiple proportions

# Overview

Review of the big pictures: Inference using computation and mathematical methods

Review and practice of parametric interference for means

Hypothesis tests for more than two proportions

- Randomization test using the chi-square statistic
- Parametric chi-square statistic test for goodness-of-fit

# Announcement: homework 10

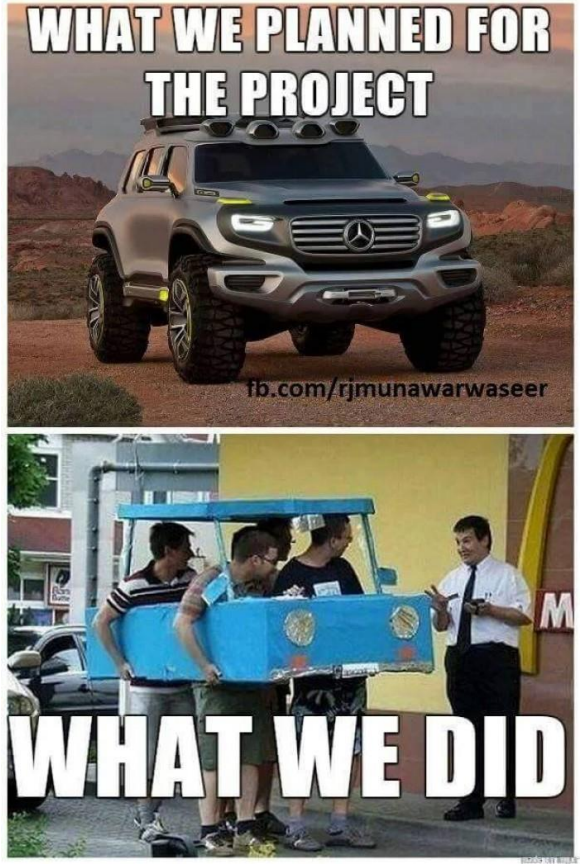
Homework 10 has been posted!

- The last homework ☹

It is due on Gradescope on Sunday April 19th at 11pm

Also, please work on your class final project

- It is due on Gradescope on Sunday April 26th



# Review: The big picture of statistical inference

Statistic inference always involves using samples of data to make claims about a population

- i.e., using statistics to make claims about parameters

To quantify the **size of an effect**, we have created <u>confidence intervals</u>

- These intervals contain a parameter of interest most of the time

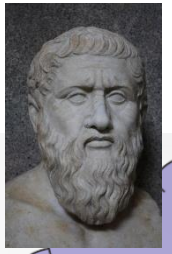
To assess if a particular **parameter value is plausible**, we have run <u>hypothesis tests</u>

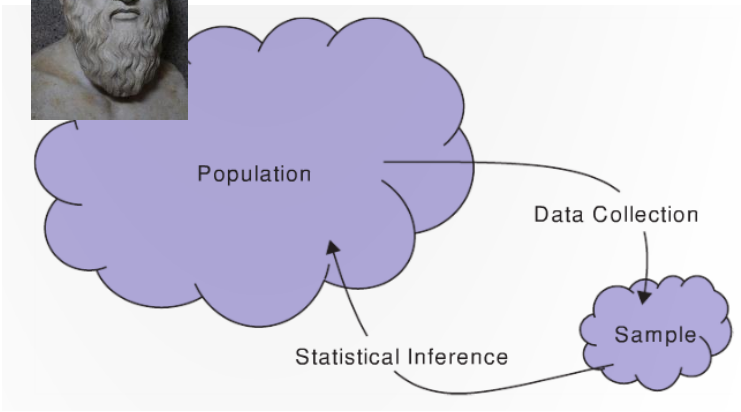
$H_0$: $\rho = 0$

$\pi, \mu, \sigma, \rho$

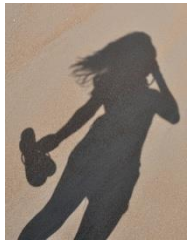
0.4 0.6

$\rho$





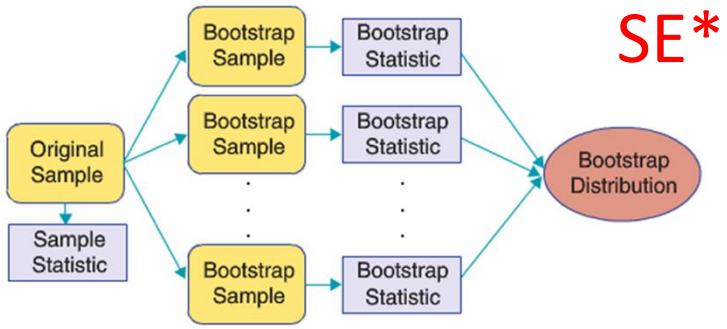

$\hat{p}, \bar{x}, s, r$



# Inference using randomization/computational methods

In the first half of the class we used computational (randomization) methods for inference

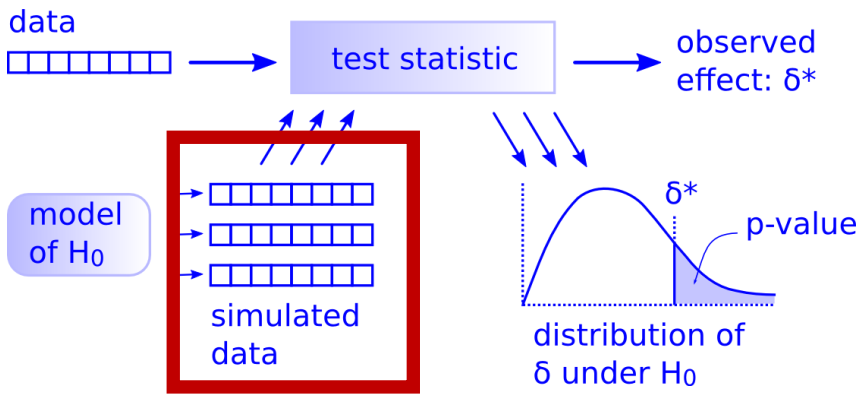
- i.e., we computationally repeat a process many times using the do_it() function




For confidence intervals, we used the bootstrap to repeatedly sample with replacement from our original sample

$$\text{CI} = \text{statistic} \pm q^* \cdot SE^*$$

For hypothesis tests, we have created null distributions by repeatedly generating statistics consistent with the null hypothesis



# Inference using randomization/computational methods

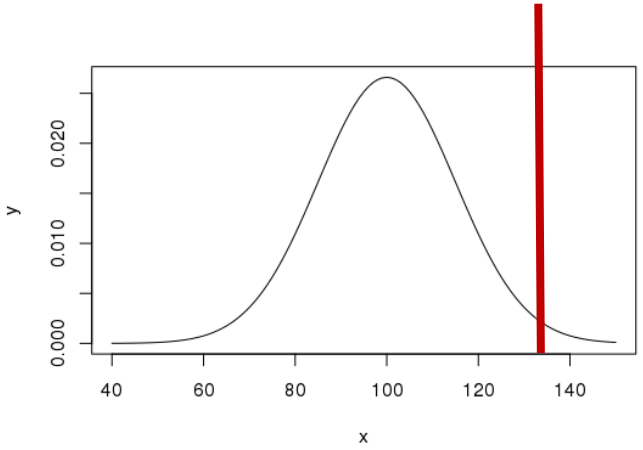
| Data | 1 Sample | 2 Samples | > 2 Samples |
|---|---|---|---|
| Categorical data | $H_0$: $\pi = p_0$<br>$H_A$: $\pi \neq p_0$<br><br><u>Flip "coins"</u><br><br>rflip() | $H_0$: $\pi_1 = \pi_2$<br>$H_A$: $\pi_1 \neq \pi_2$<br><br><u>Flip "coins"</u><br><br>rflip() | $H_0$: $\pi_1 = p_1$, $\pi_2 = p_2$, … , $\pi_k = p_k$<br>$H_A$: At least one $p_i$ is different than specified<br><br><u>Roll a k-sided die n times</u><br><br>rroll(n, prob = ) |
| Quantitative data | $H_0$: $\mu = v_0$<br>$H_A$: $\mu \neq v_0$<br><br><u>resample</u><br><br>sample(… , replace = TRUE) | $H_0$: $\mu_1 = \mu_2$<br>$H_A$: $\mu_1 \neq \mu_2$<br><br><u>Shuffle data</u><br><br>shuffle() | $H_0$: $\mu_1 = \mu_2 = \ldots = \mu_k$<br>$H_A$: At least one $\mu_i$ is different<br><br><u>Shuffle data</u><br><br>shuffle() |

# Inference parametric probability models

In the past two weeks we used mathematical functions instead of computational methods

For confidence intervals, we will have formulas for SE estimates

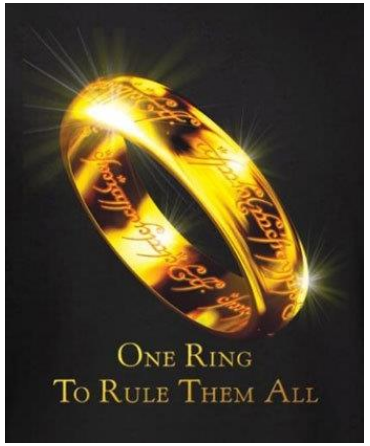
- CI = stat ± q* · $\hat{SE}$ $\qquad \hat{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$




For hypothesis tests, we will have mathematical functions for the null distribution

- E.g., our null distribution will be a normal distribution

But conceptually, everything is the same!



| Data | 1 Sample | 2 Samples | > 2 Samples |
|---|---|---|---|
| Categorical data | $H_0$: $\pi = p_0$<br>$H_A$: $\pi \neq p_0$<br><br><u>z-test</u><br><br>$z = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$ | $H_0$: $\pi_1 = \pi_2$<br>$H_A$: $\pi_1 \neq \pi_2$<br><br><u>z-test</u><br><br>$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}}$ | $H_0$: $\pi_1 = p_1$, $\pi_2 = p_2$, … , $\pi_k = p_k$<br>$H_A$: At least one $p_i$ is different than specified<br><br><u>chi-square test</u><br><br>$\chi^2 = \sum_{i=1}^{k} \frac{(Observed_i - Expected_i)^2}{Expected_i}$<br><br>df = k - 1 |
| Quantitative data | $H_0$: $\mu = v_0$<br>$H_A$: $\mu \neq v_0$<br><br><u>One sample t-test</u><br><br>$t = \frac{\bar{x}-v_0}{s/\sqrt{n}}$<br><br>df = n - 1 | $H_0$: $\mu_1 = \mu_2$<br>$H_A$: $\mu_1 \neq \mu_2$<br><br><u>Two sample t-test</u><br><br>$t = \frac{\bar{x}_1-\bar{x}_2}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}}$<br><br>df = min $n_1$ - 1, $n_2$ - 1 | $H_0$: $\mu_1 = \mu_2 = \ldots = \mu_k$<br>$H_A$: At least one $\mu_i$ is different<br><br><u>Analysis of Variance</u><br><br>$F = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$<br><br>$df_1 = k - 1$, $df_2 = n - k$ |

# Computational test

Are more than 10% of lawyers left-handed?

1. State the hypotheses

   $H_0$: $\pi = 0.1$

   $H_A$: $\pi > 0.1$

2. Calculate the observed statistic

   16 of 105 were left-handed

   $\hat{p} = 16/105 = .152$

3. Create a null distribution

   rflip(105, .1)/105

4. Calculate the p-value

   pnull($\hat{p}$, null_dist, lower.tail = FALSE)

5. Make a judgement

# Parametric test

1. State the hypotheses

   $H_0$: $\pi = 0.1$

   $H_A$: $\pi > 0.1$

$$z = \frac{\hat{p} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$$

2. Calculate the observed statistic

   z = (.152 - .1)/sqrt((.1 * (1 - .1))/105)

3. Visualize the null distribution

   dnorm(x_vals, 0, 1)

4. Calculate the p-value

   pnorm(z, 0, 1, lower.tail = FALSE)

5. Make a judgement

Usually, should come to the same conclusion!

# Computational test

Is this pill effective?

1. State the hypotheses

   $H_0$: $\mu_T = \mu_C$

   $H_A$: $\mu_T > \mu_C$

2. Calculate the observed statistic

   $\bar{x}_{Effect} = \bar{x}_{Treatment} - \bar{x}_{Control}$

   $\bar{x}_{Effect} = 5 - -.272 = 5.272$

3. Create a null distribution

   shuffle(combined_data)

4. Calculate the p-value

   pnull($\bar{x}_{Effect}$ , null_dist, lower.tail = F)

5. Make a judgement

# Parametric test

1. State the hypotheses

   $H_0$: $\mu_T = \mu_C$

   $H_A$: $\mu_T > \mu_C$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

2. Calculate the observed statistic

   t = 5.272/SE = 1.60

3. Visualize the null distribution

   dt(x_vals, df = 9)

4. Calculate the p-value

   pt(t, df = 9, lower.tail = FALSE)

5. Make a judgement

Usually, should come to the same conclusion!

# Plan for the rest of the semester

Review/practice on parametric inference on means

Additional hypothesis tests:

- Randomization and parametric tests for more than two proportions
- Parametric tests for more than two means

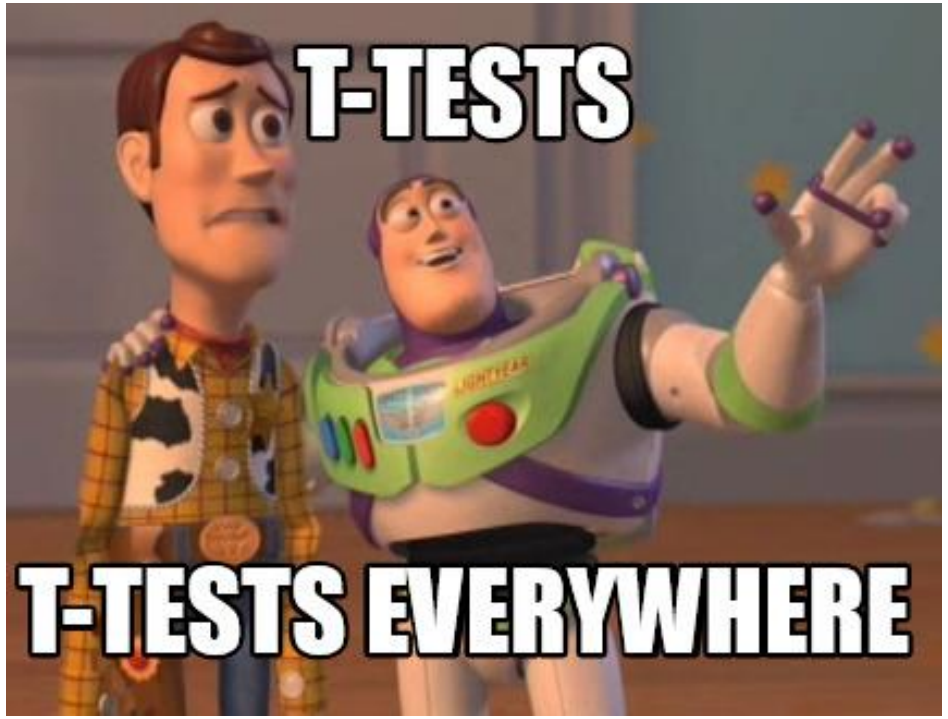
Theories of hypothesis test

Inference for simple linear regression (β)

Brief discussion of multiple regression

- i.e., regression with more than one explanatory variable)

# Review: parametric inference for means

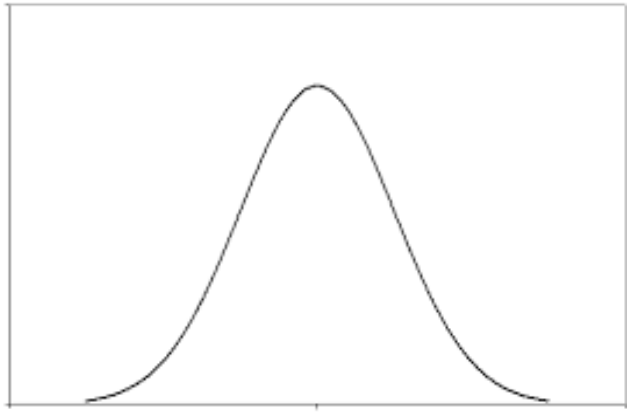


# Review: Using parametric methods for inference

When using “parametric” methods for statistical inference, we have mathematical functions for our distributions of statistics

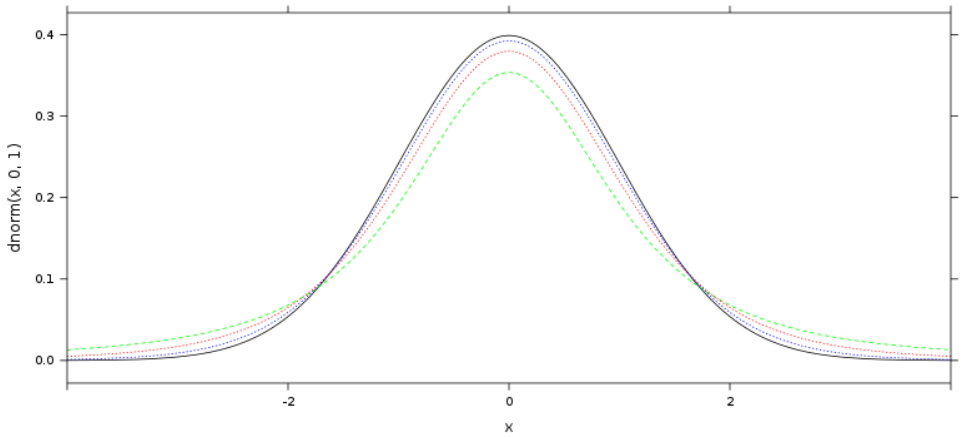
- e.g., we have mathematical functions for the null distribution rather than needing to use randomization methods such as shuffling our data

Normal distribution N(μ,σ)



Inference on a single proportion π

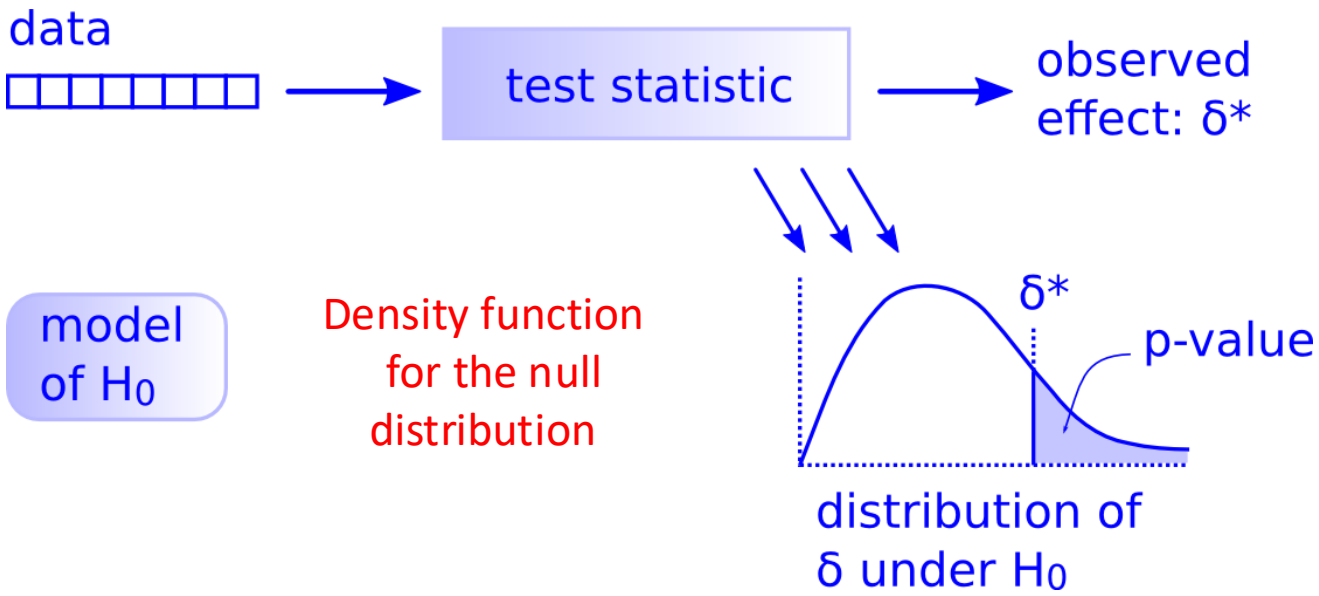
T-distribution

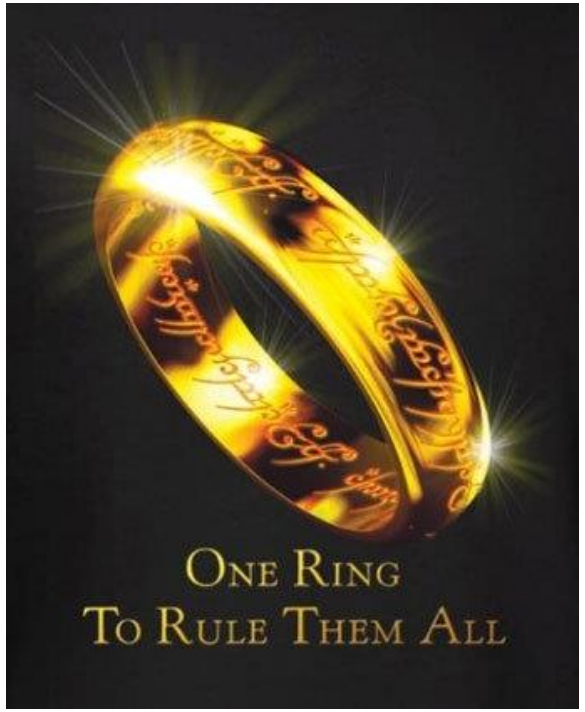



Inference on means μ (when σ is unknown)

# One test to rule them all

There is only one hypothesis test!

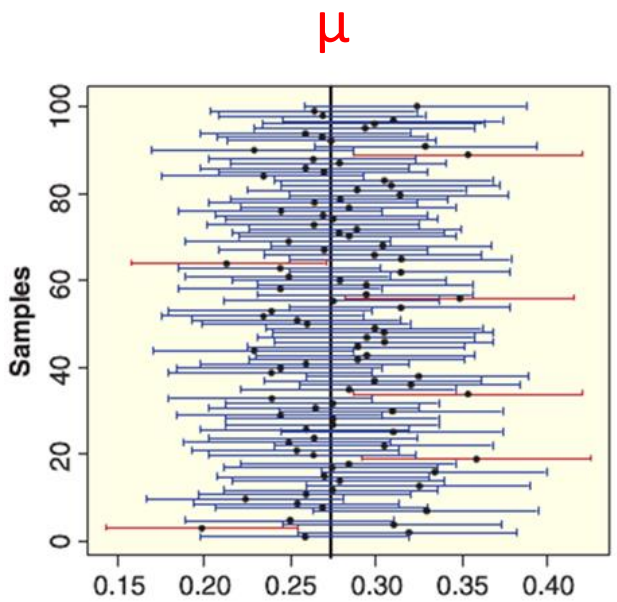



Just follow the 5 hypothesis tests steps!

CI: stat ± $q^* \cdot SE$

# Confidence intervals for means




**Single mean μ:** $\bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$

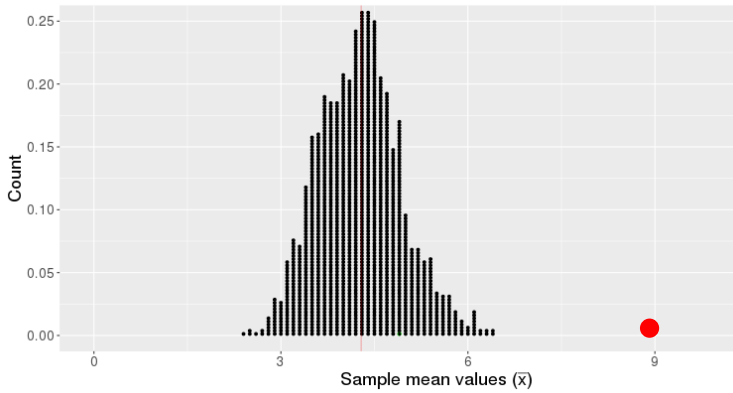
**Difference of two means $\mu_1$ - $\mu_2$:** $(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

**Mean of differences for paired data $\mu_d$:** $\bar{x}_d \pm t^* \cdot \frac{s_d}{\sqrt{n_d}}$

(the t-distribution is appropriate provided n is large (≥ 30) or the data is not too skewed)

# Hypothesis tests means (t-tests)

$H_0$: ~~$\mu$~~ = $\mu_0$




**Single mean μ:** $$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$ df = n - 1

- $H_0$: $\mu = \mu_0$

**Comparing two means $\mu_1 - \mu_2$:** $$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$ df = min($n_1$, $n_2$) - 1

- $H_0$: $\mu_1 = \mu_2$

**Comparing mean of differences for paired data $\mu_d$:** $$t = \frac{\bar{x}_d}{\frac{s_d}{\sqrt{n}}}$$ df = n - 1

- $H_0$: $\mu_d = 0$

(the t-distribution is appropriate provided n is large (≥ 30) or the data is not too skewed)

# Warm-up problem 1: Are you getting enough sleep?

It is recommended that adults sleep at least 8 hours a night

A Statistics professor asked 12 undergraduate students how much sleep they were getting and found the average was 6.2 hours with a standard deviation of 1.7 hours

Assuming this is representative of all students in a Statistics class, does this provide evidence that students in the class are not getting enough sleep on average?

Use R to run hypothesis test to examine this question!

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}}$$

# Getting enough sleep?

What are the null and alternative hypotheses?

- $H_0$: $\mu = 8$
- $H_A$: $\mu < 8$

What are the values of the following?

- $\bar{x} = 6.2$
- $n = 12$
- $s = 1.7$

Plugging in to our formula we get:

- t = (6.2 - 8)/(1.7/sqrt(12)) = -3.67
- pt(-3.67, 11) = 0.0018

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

# Quick review: paired t-test

# Do left or right tires wear out quicker?

| Car # | Right tire | Left tire |
|---|---|---|
| 1 | 42 | 54 |
| 2 | 75 | 73 |
| 3 | 24 | 22 |
| 4 | 56 | 59 |
| 5 | 52 | 51 |
| 6 | 56 | 45 |
| 7 | 23 | 29 |
| 8 | 55 | 58 |
| 9 | 46 | 49 |
| 10 | 52 | 58 |
| 11 | 47 | 49 |
| 12 | 62 | 67 |
| 13 | 55 | 58 |
| 14 | 62 | 64 |

# Do left or right tires wear out quicker?

**Independent samples t-test**

$H_0$: $\mu_{left} - \mu_{right} = 0$

$H_A$: $\mu_{left} - \mu_{right} \neq 0$

$$t = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

We will use the minimum of $n_1 - 1$, or $n_2 - 1$ as a conservative estimate of the df

| Car # | Right tire | Left tire |
|---|---|---|
| 1 | 42 | 54 |
| 2 | 75 | 73 |
| 3 | 24 | 22 |
| 4 | 56 | 59 |
| 5 | 52 | 51 |
| 6 | 56 | 45 |
| 7 | 23 | 29 |
| 8 | 55 | 58 |
| 9 | 46 | 49 |
| 10 | 52 | 58 |
| 11 | 47 | 49 |
| 12 | 62 | 67 |
| 13 | 55 | 58 |
| 14 | 62 | 64 |

# Do left or right tires wear out quicker?

**Paired samples t-test**

$H_0$: $\mu_{diff} = 0$

$H_A$: $\mu_{diff} \neq 0$

$$t = \frac{\overline{x}_d}{s_d / \sqrt{n_d}}$$

The t-statistic comes from a t-distribution with n - 1 degrees of freedom

Should we try it in R?

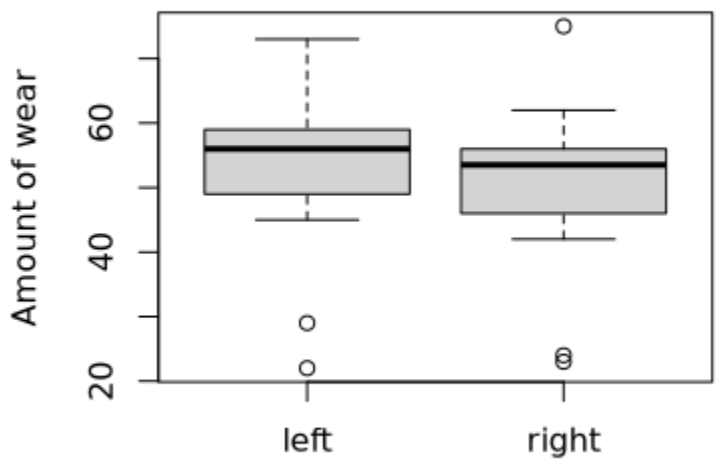
| Car # | Right tire | Left tire |
|---|---|---|
| 1 | 42 | 54 |
| 2 | 75 | 73 |
| 3 | 24 | 22 |
| 4 | 56 | 59 |
| 5 | 52 | 51 |
| 6 | 56 | 45 |
| 7 | 23 | 29 |
| 8 | 55 | 58 |
| 9 | 46 | 49 |
| 10 | 52 | 58 |
| 11 | 47 | 49 |
| 12 | 62 | 67 |
| 13 | 55 | 58 |
| 14 | 62 | 64 |

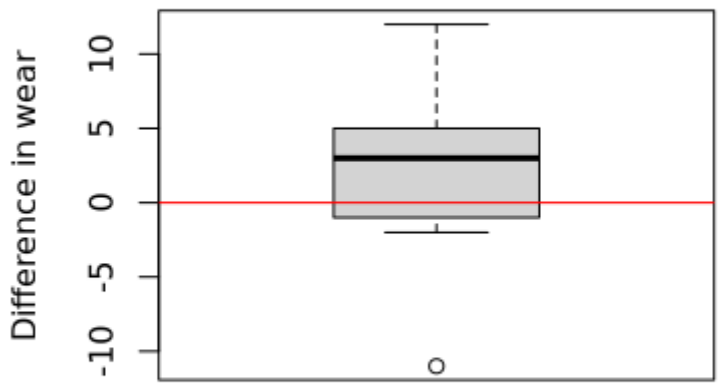
# R code for tire wear

```
car_tires <- read.csv("car_tires.csv")
left_tire <- car_tires$left
right_tire <- car_tires$right

boxplot(left_tire, right_tire,
        names = c("left", "right"))

tire_diff <- left_tire - right_tire
boxplot(tire_diff)
abline(h = 0, col = "red")
```



# R code for tire wear

```
t.test(left_tire, right_tire)                    # p-value = 0.6958

t.test(left_tire, right_tire, paired = TRUE)     # p-value =  0.1618
```

The paired t-test is more “powerful” test

- i.e., has better chance of rejecting $H_0$ (when $H_0$ is false)

# Randomization test to compare multiple proportions

# Testing more than two categories

A Yale professor was interesting in examining whether Yale students were equally likely to be born in each month

$Q_1$: What is the null and alternative hypotheses he could use to test this hypothesis?

# Testing more than two categories

The hypotheses:

$H_0$: $\pi_{Jan} = \pi_{Feb} = \ldots = \pi_{Dec} = 1/12$

$H_A$: At least one of the proportions $\pi_i$ is different

# Chi-square goodness-of-fit test

If we want to test proportions for k > 2 categories, we can use:

1. A randomization test
2. A parametric "*chi-square goodness-of-fit*" test

These test:

$H_0$: $\pi_1 = a$, $\pi_2 = b$, … $\pi_k = z$

$H_A$: Some $\pi_i$ is not as specified in $H_0$

The tests don't specify which proportion differs from what is specified in the null hypothesis, just that at least one proportion does differ

# Chi-square statistic

The **chi-square statistic**, denoted $\chi^2$, is found by comparing the **observed counts** from a sample with the **expected counts** derived from a null hypothesis and is computed as:

$$\chi^2 = \sum_{i=1}^{k} \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i}$$

Note this is a Greek symbol even though it is a statistic ☹

# Birth months from 198 Yale students

| | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | **Total** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Observed | 20 | 17 | 21 | 24 | 11 | 18 | 12 | 17 | 13 | 14 | 17 | 14 | **198** |

Q: What is the expected value for each month? A: $1/12 \cdot 198$

$$\chi^2 = \sum_{i=1}^{k} \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i} = 10.12$$

Q: Is the observed statistic beyond what we would expect if $H_0$ was true?

# Creating a null distribution

Any ideas how we could create a null distribution using randomization methods?

We could use randomization method:

- Roll k-sided weighted die
- Probability of getting each side is equal to the proportions specified in the null hypothesis
- Roll the die n times to simulate one experiment
  - Calculate $\chi^2$ statistic based on these rolls
- Repeat many times to get a null distribution

# Creating a null distribution

For the Yale birth month example…

How many sides would the die have?

- 12-sided die

What would the probability be of getting each side?

- 1/12

How many times would we roll the die to simulate one data set?

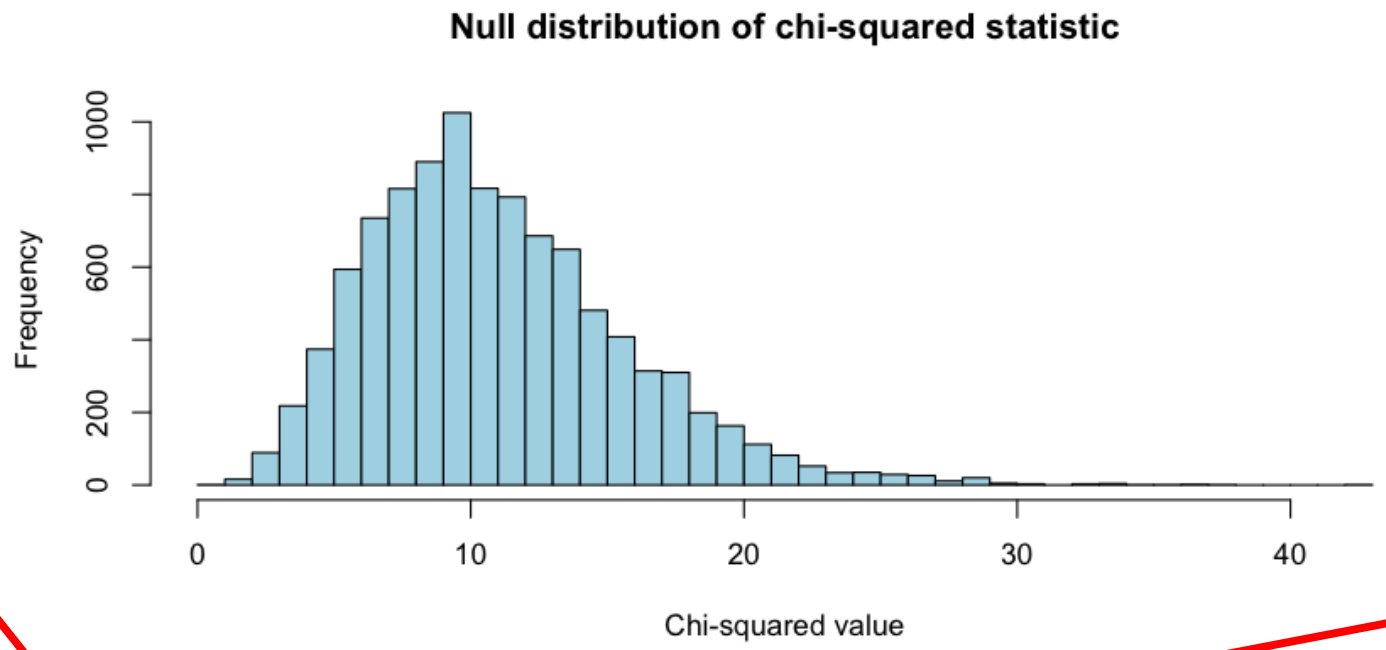
- n = 198
- Calculate $\chi^2$ statistic based on these 198 rolls

$$\chi^2 = \sum_{i=1}^{k} \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i}$$

How many times would we repeat this process?

- 10,000 times

# Randomization null distribution




Vector of randomly generated counts in each category

"observed" data consistent with $H_0$

Vector of expected proportions for each category level

Vector length is the number of categories k

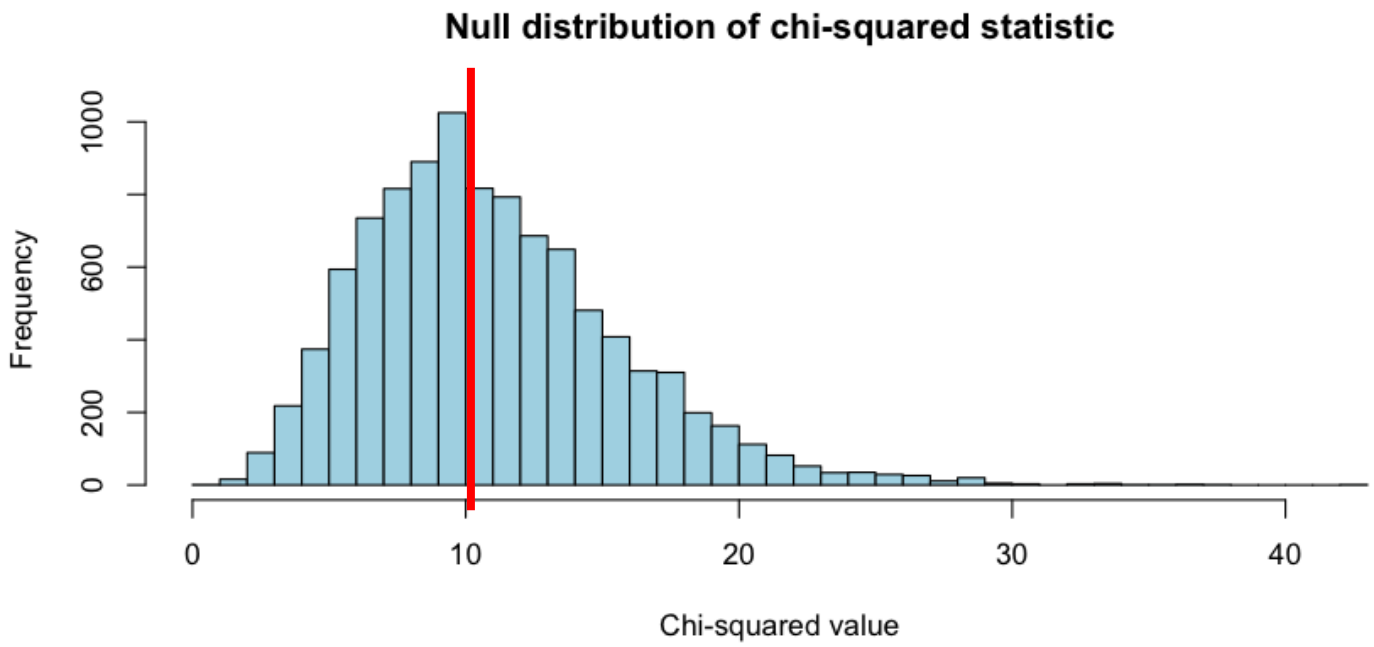
```
simulated_counts <- rroll(n,  expected_proportions)

simulated_counts <- rroll(198,  rep(1/12, 12))
```

Calculate statistic $\chi^2$ and repeat 10,000 times

$$\chi^2 = \sum_{i=1}^{k} \frac{(\text{Observed}_i - \text{Expected}_i)^2}{\text{Expected}_i}$$

# Randomization null distribution

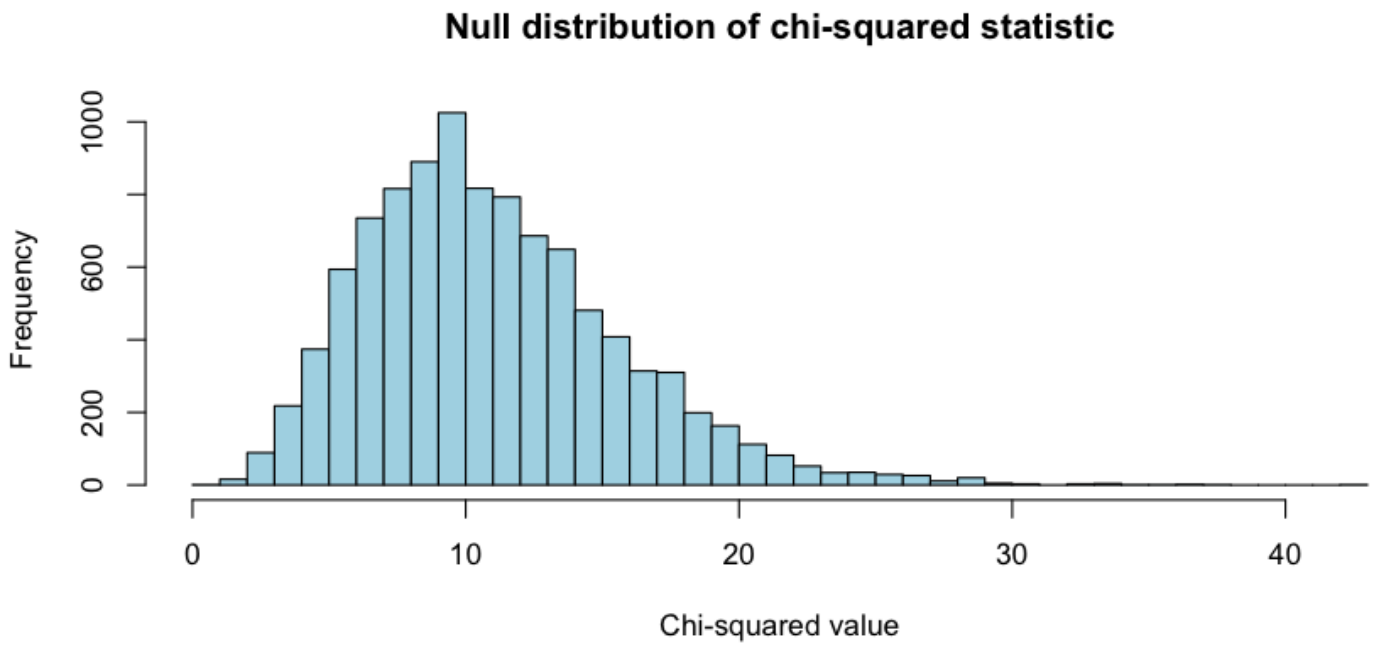



$\chi^2 = 10.12$

p-value = …?

# $\chi^2$ test for goodness-of-fit to compare multiple proportions

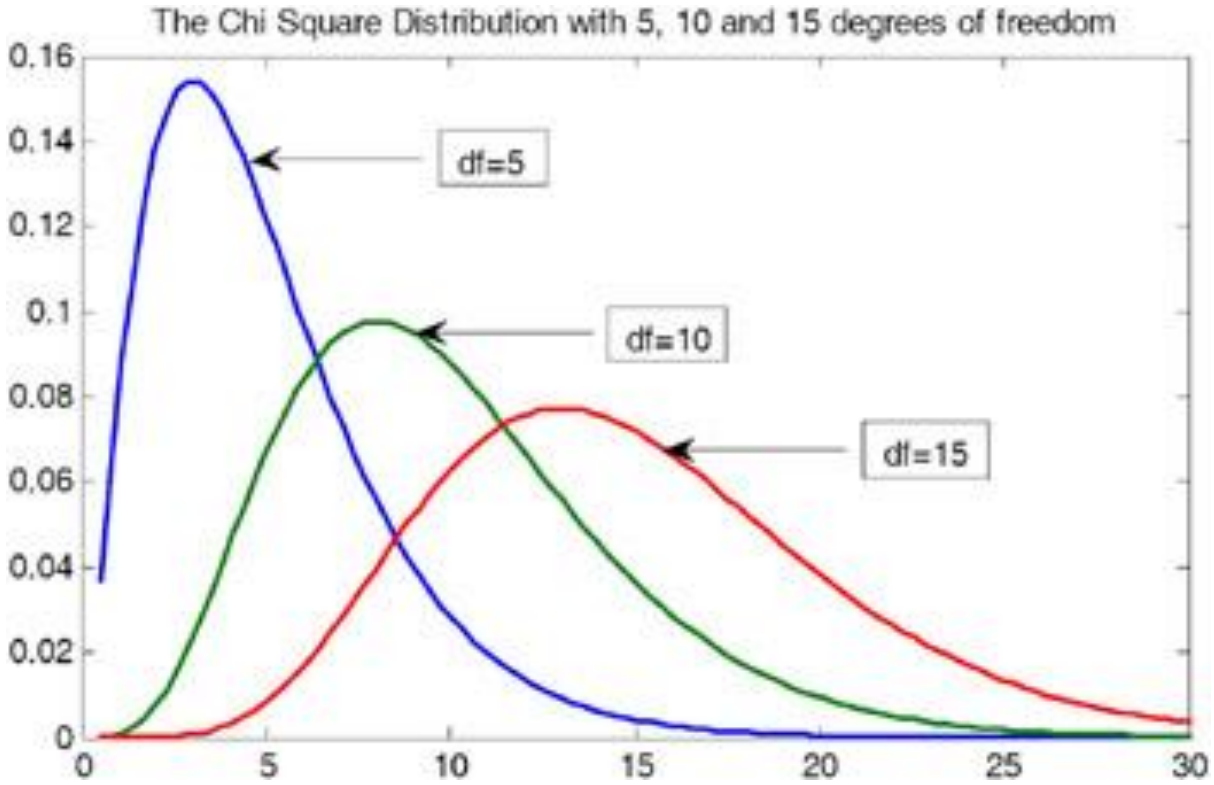
# Parametric null distribution




Is there a parametric null distribution we could use instead of simulations?

Yes! The $\chi^2$ distribution!

# Chi-square distribution

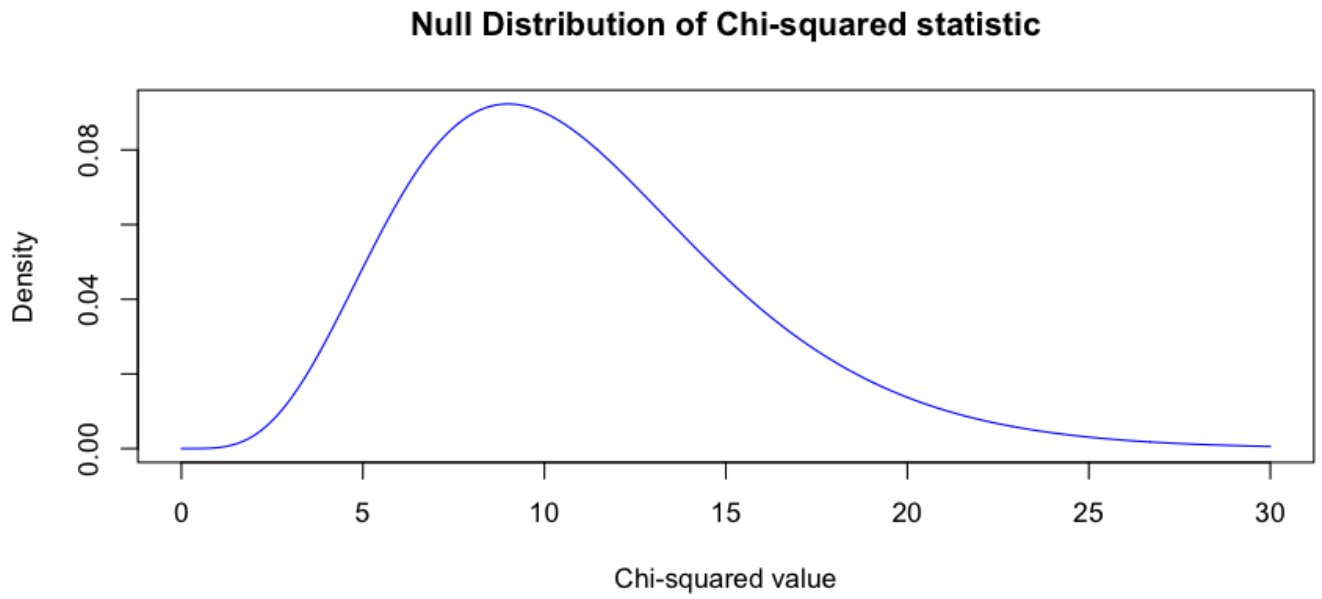



k is the number of groups

The $\chi^2$ has one parameter called 'degrees of freedom', which is equal to k - 1

$\chi^2$ distribution can be used as a null distribution as long as there are at least 5 expected counts in every condition

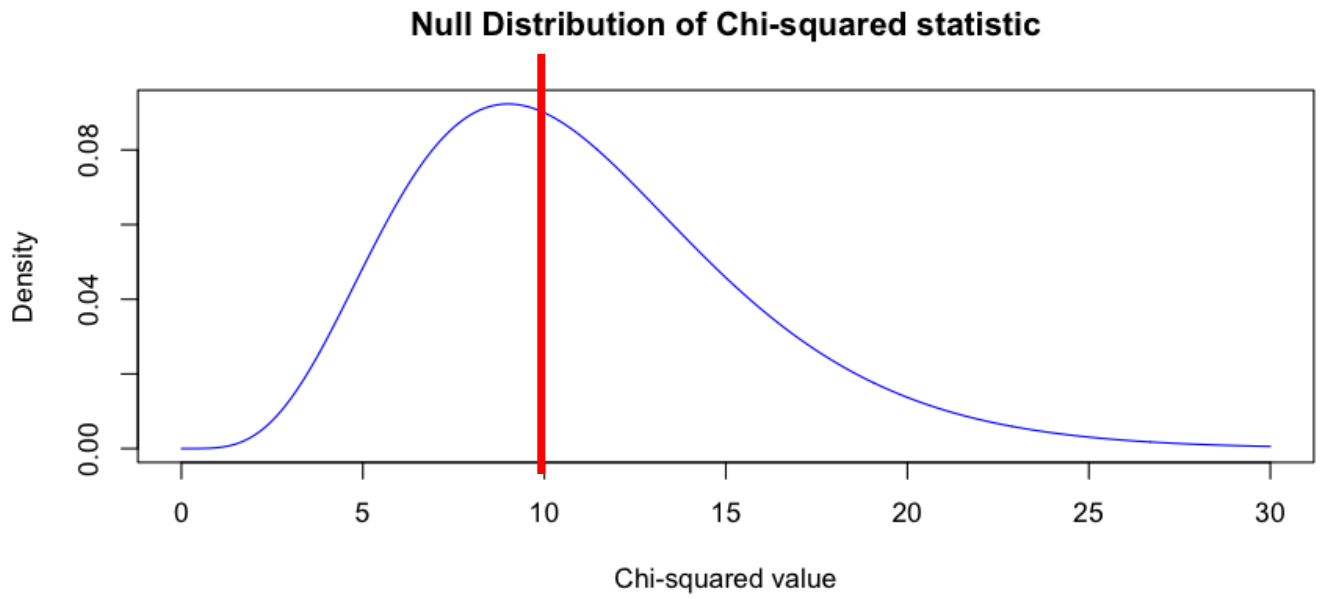
# Chi-square distribution




To plot the chi-squared density we can use: y_vals <- dchisq(x_vals, df)

To get a p-value we can use: pchisq(chi_stat, df, lower.tail = FALSE)

# Chi-square distribution




For Yale birth month example, since there 12 months there are 11 degrees of freedom (df = 11)

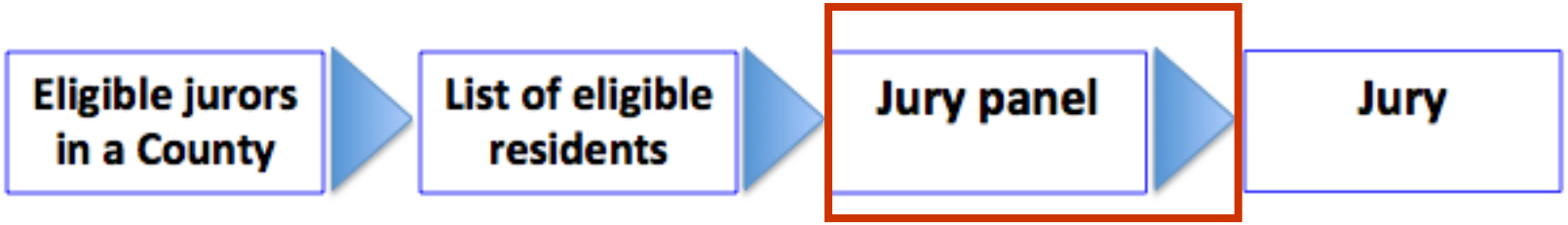
P-value based on the chi-squared distribution = …?

# Another example

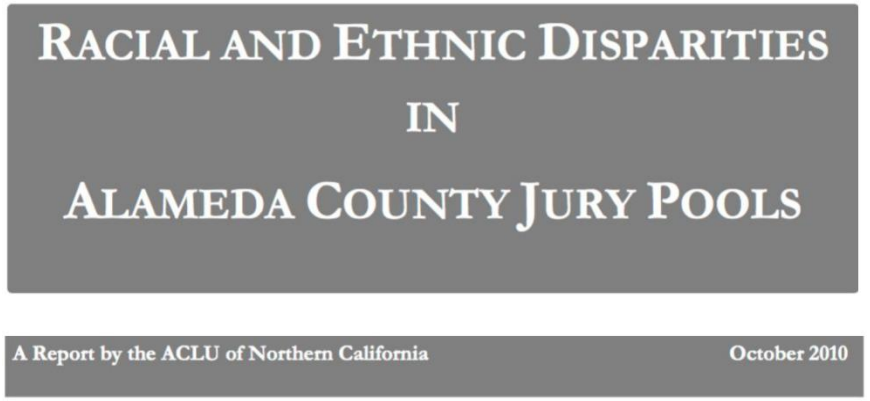
# Jury selection in Alameda county

Section 197 of California's Code of Civil Procedure says:

" All persons selected for jury service shall be selected at random, from a source or sources inclusive of a representative cross section of the population of the area served by the court."




In 2010, the American Civil Liberties Union (ACLU) of Northern California presented a report that concluded that certain racial and ethnic groups are underrepresented among jury panelists in Alameda County.



# Jury selection in Alameda county data

The ACLU compiled data on the composition of 1453 people who were on jury panels from in the years 2009 and 2010.

The demographics and jury panel proportions were:

| | Asian | Black | Latino | White | Other |
|---|---|---|---|---|---|
| Eligible | 0.15 | 0.18 | 0.12 | 0.54 | 0.01 |
| Panel | 0.26 | 0.08 | 0.08 | 0.54 | 0.04 |

**Question**: were ethnicities selected to be on the panel representative of what would be expected from the underlying population?

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**Question**: What are the null and alternative hypotheses?

$H_0$: $\pi_A = .15$, $\pi_B = .18$, $\pi_L = .12$, $\pi_W = .54$, $\pi_o = .01$

$H_A$: Some $\pi_i$ is not as specified in $H_0$

Let's try it in R!